

CS 6603: AI, Ethics, and Society

Homework Project #4: AI/ML Part 2

Readings:

- Chapter 7: Weapons of Math Destruction (Sweating Bullets: On the Job)
- “A Few Useful Things to Know about Machine Learning” by Pedro Domingos
<https://homes.cs.washington.edu/~pedrod/papers/cacml12.pdf>

In this assignment, you’ll apply AI/ML algorithms to two applications: word embeddings and facial recognition.

Task 1: Word2vec

In this task, you will use distributional vectors trained using Google’s deep learning Word2vec system.

1. Familiarize yourself with word2vec.
 - (a) Read the original paper on word2vec - [Mikolov et al. \(2013\)](http://papers.nips.cc/paper/5021-distributed-representations-of-words-and-phrases-and-their-compositionality.pdf) (<http://papers.nips.cc/paper/5021-distributed-representations-of-words-and-phrases-and-their-compositionality.pdf>).
 - (b) Learn about the system and how to train your own vectors. More information [here](https://code.google.com/archive/p/word2vec/) (<https://code.google.com/archive/p/word2vec/>).
 - (c) Learn about the python wrapper around Word2vec [here](https://rare-technologies.com/word2vec-tutorial/). (<https://rare-technologies.com/word2vec-tutorial/>)
2. Install [Gensim](https://pypi.org/project/gensim/) on your machine.
 - (a) Example: `pip install gensim | pip install --upgrade gensim`
3. Download the pre-trained Word2vec model.
 - (a) The file (`reducedvector.bin`) for the provided model can be downloaded from Canvas (under the AI/ML Assignment Part 2).
 - (b) Place this file within your project directory.
 - (c) For more information about this model (created based on the Google News dataset), see <https://code.google.com/archive/p/word2vec/>
4. To use the Gensim and the pre-trained Word2vec model, use the following code to create the model that will be used in your project:

```
from gensim.models import Word2Vec
import gensim.models
import nltk
newmodel = gensim.models.KeyedVectors.load_word2vec_format(<path to
reducedvector.bin>, binary=True)
```
5. In the project you will compute similarity measures associated with words within the model. For example, to find different measures of similarity based on the data in the Word2vec model, we can use:

```
# Find the five nearest neighbors to the word man
newmodel.most_similar('man', topn=5)

# Compute a measure of similarity between woman and man
newmodel.similarity('woman', 'man')
```
6. In the project you will compute analogies. Analogies like man is to woman as king is to __?__, can be computed using:

```
newmodel.most_similar(positive=['king', 'woman'], negative=['man'],
                      topn=1)
```

Based on your understanding of the Word2vec system, answer the following questions in your report.

1.1: Use Word2vec to compute bias in target words *man* and *woman*

For both target words *man* and *woman*, use the pre-trained Word2vec model to order the following 15 words from **most similar to least similar**. For each word-target word pair, provide the similarity score. Provide your results in table format.

wife
husband
child
queen
king
man
woman
birth
doctor
nurse
teacher
professor
engineer
scientist
president

1.2: Word2vec and the Bigger Analogy Test Set (BATS)

For this question, you will use the Bigger Analogy Test Set (BATS) Word analogy task. This has been one of the standard benchmarks for word embeddings since 2013 (<https://vecto.space/projects/BATS/>).

Part A: Using Word2vec on a file from the Bigger Analogy Test Set (BATS)

- (a) Download the `BATS_3.0.zip` dataset from Canvas (under AI/ML Assignment Part 2).
- (b) Select one file from the dataset. In your report, note the file selected.
- (c) For each row in your selected file:
 - a. Choose one word as the target word
 - b. Compute the measure of similarity between the target word and each of the other words in the row.
 - c. If the row contains more than five additional words (excluding the target word), report similarity scores for any five words in the row.
 - d. Document the results in a table format.

Part B: Analyzing Bias in Word Embeddings Among Members of a Protected Class

- (a) Choose a protected class.
- (b) Choose three words that identify membership in the selected protected class (ex: Young and Old are members of the protected class Age).
- (c) Provide the following in table format:
 - a. For each row in your selected `BATS_3.0` file, compute the similarity between your target word and each of your three words.
 - b. Include a column for noticeable differences in the similarity scores based on membership in the protected class.

Note: What constitutes a “noticeable difference” is left as an exercise for the student to decide.

- (d) Based on the noticeable differences identified in **Step (c)**, explain why these differences in similarity scores may exist among words representing members of the protected class. Did you observe any patterns that suggest potential patterns of bias?

1.3: Computing analogies with Word2vec

For this question you will use the following sentences:

king is to throne as judge is to ____?
giant is to dwarf as genius is to ____?
college is to dean as jail is to ____?
arc is to circle as line is to ____?
French is to France as Dutch is to ____?
man is to woman as king is to ____?
water is to ice as liquid is to ____?
bad is to good as sad is to ____?
nurse is to hospital as teacher is to ____?
usa is to pizza as japan is to ____?
human is to house as dog is to ____?
grass is to green as sky is to ____?
video is to cassette as computer is to ____?
universe is to planet as house is to ____?
poverty is to wealth as sickness is to ____?

- (a) Complete the above sentences with your own word analogies (i.e., think of a word to fill in the blank). Then, use Word2vec to find the similarity score between the pair of words in the second part of each analogy. In your report, provide the word that answers each analogy and the similarity score.

For Example:

man is to woman as king is to queen?
newmodel.similarity('king', 'queen') -> 0.5685571

- (b) Use the Word2vec model to find the word analogy and corresponding similarity score. In your report, provide the word that answers each analogy and the similarity score.

For Example:

man is to woman as king is to ____?
newmodel.most_similar(positive=['king', 'woman'], negative=['man'], topn=1) -> queen,
0.711

- (c) Lastly, compute and print the correlation between the vector of similarity scores from your analogies versus the Word2vec analogy-generated similarity scores. What is the strength of the correlation?
- .00-.19 “very weak” correlation
 - .20-.39 “weak” correlation
 - .40-.59 “moderate” correlation
 - .60-.79 “strong” correlation
 - .80-1.0 “very strong” correlation

Note: There will be a single value for the correlation.

- (d) Using the correlation calculated in Step c, what can you infer about how well your predictions line up with the model predictions?

Task Set 2: UTK Dataset

In this task, we will work with a UTK dataset. This dataset is a subset of the UTKFace dataset, which contains over 20,000 images of faces annotated with age, gender, and ethnicity. Information about the original UTKFace dataset can be found at <https://susanqq.github.io/UTKFace/>.

- (a) Download the **UTKface_cropped.tar.gz** from Canvas (under the AI/ML Assignment Part 2).

2.1 Analysis of UTK Dataset

Each image in the dataset has a unique value representing age, gender, and race based on the following legend:

- age: indicates the age of the person in the picture and can range from 0 to 116.
- gender: indicates the gender of the person and is either 0 (male) or 1 (female).
- race: indicates the race of the person and can range from 0 to 4, denoting White, Black, Asian, Indian, and Others (like Hispanic, Latino, Middle Eastern).

Please complete and answer the following:

- (a) Compute and document the frequency of images associated with each subgroup for age (subdivide based on - (0-20), (21,40), (41,60), (61,80), (81, 116)), gender (0,1), and race (0 to 4).
- (b) Using the frequency counts, recreate a table of the age group, gender, and race distributions of subjects based on the UTK dataset subgroups. Please see the table below as an example.

Age group	0-20	21-40	41-60	61+	Total
Female	1,248	1,685	1,011	165	4,109
Male	1,427	2,501	5,021	2,641	11,590
Black	40	532	354	219	1,145
White	1,497	3,368	5,140	2,368	12,373
Asian	1,126	284	537	219	2,166
Unknown	12	2	1	0	15
Total	2,675	4,186	6,032	2,806	15,699

http://biometrics.cse.msu.edu/Publications/Face/HanJain_UnconstrainedAgeGenderRaceEstimation_MSUTechReport2014.pdf

- (c) In your report, answer the following questions:
- Based on your calculations for age, which subgroup has the largest representation? Which subgroup has the least representation?
 - Based on your calculations for gender, which subgroup has the largest representation? Which subgroup has the least representation?
 - Based on your calculations for race, which subgroup has the largest representation? Which subgroup has the least representation?
- (d) Based on what we've learned so far, if an algorithm is trained based on this dataset, which group(s) will be impacted the most? Why? Provide a real-world example where a model trained on this dataset might cause harm. Please discuss your answer in a minimum of 1 - 2 paragraphs with supporting information from the class lectures.

Task 3: Submission

Submit a PDF report documenting your outputs for each question.

- **The report must follow JDF formatting.**
 - You can find a link to the JDF template here: [JDF Templates](#). The Microsoft Word template is recommended to ensure proper formatting and styling.
 - Reports that do not follow JDF formatting, or that are not neat and well organized, may receive up to a 10% deduction, regardless of content quality.
- **Name your file exactly as follows: GTuserName_Assignment_4.pdf**
 - *Example: gBurdell3_Assignment_4.pdf*
 - Deductions may be applied if the file name is incorrect or does not follow this convention.
- **All charts, graphs, and tables must be generated using Python, Excel, or another suitable non-AI software application.**
 - Points may be deducted for charts or tables generated using AI tools.
- If you used a Jupyter Notebook (.ipynb) for your work, you may submit it as a supplementary file. However, note that **PDF report is required** and will be used for grading.